

FAZ II Classic.
The cost-efficient fixing
for high performance
requirements in
cracked concrete.



FAZ II Classic.

For high performance requirements.

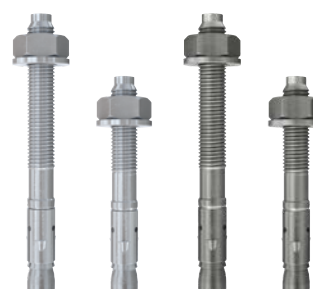
The distinctive edge holds the expansion clip in position even with reinforcement hits, thus ensuring **safe installation**.



The interaction of cone and expansion clip significantly increases load-bearing capacity and makes **minimum edge distance and axial spacing possible**.

Assortment

The FAZ II Classic is available in galvanised and stainless steel (R) as well as in a short version (K).



Approvals



ETA-23/0162 of 2023/03/14 for cracked concrete and non cracked concrete



Fire resistance classification R120
Anchor available in version 1.4401

Functioning

- The FAZ II Classic is suitable for pre- and push-through mounting and also ideal for distance mounting.
- The shallow drill hole depth of the K version enables even faster installation and reduces the number of reinforcement hits.
- The ETA (assessment) also covers the use of hollow drills and diamond core bits.

Building materials

Suitable for building materials, such as



Cracked concrete



Uncracked concrete



Available software

For an easy and fast usage the anchor can be calculated with its load data in the software FiXperience.

Download-Link: www.fischer-international.com/en/service/planning-aids/fixperience

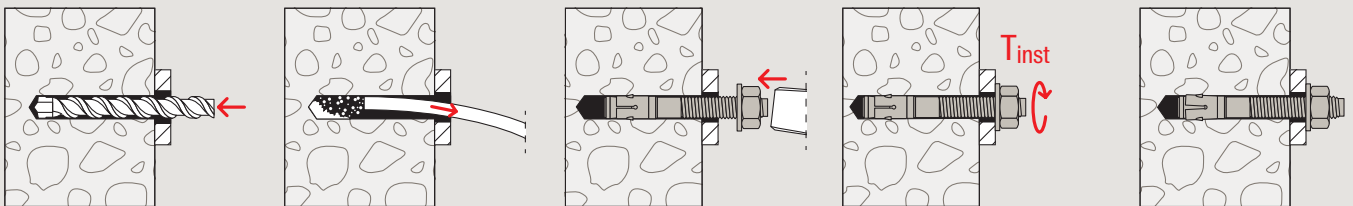
Advantages, installation and applications.

Your advantages at a glance

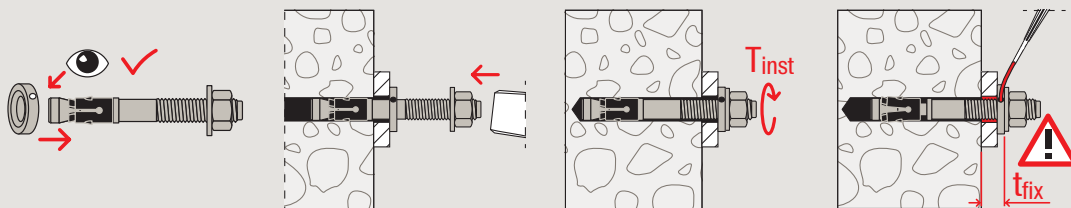
- The ETA approval Option 1 and fire resistance approval R120 in cracked and non-cracked concrete ensure a high quality standard and enable all safety-relevant applications.
- The full range M8-M16 in galvanised and stainless steel allows a wide field of indoor and outdoor applications.
- Thanks to the seismic approval (C1 and C2), the anchors can also be used safely in earthquake zones.
- The short version is particularly suitable for avoiding reinforcement hits due to the low anchoring depth.
- Quick and easy installation is possible with the help of the FA-ST II setting tool.
- The two possible anchorage depths offer more flexibility in the application.

Installation

Installation FAZ II Classic push-through installation



Installation FAZ II Classic Seismic



Application



Facade substructure



High-bay warehouse

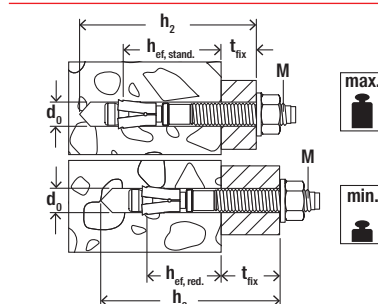


Cable trays



Stair railings

Assortment



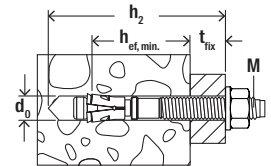
Bolt anchor FAZ II Classic



FAZ II Classic

	Galvani- sed steel	Stainless steel	Seismic- Approval	Drill diameter	Min. drill hole depth for through fixings	Max. usable length $h_{ef,stand.}/h_{ef,min.}$	Anchor length	Thread	Installation torque	Width across nut	Sales unit
	Item No.	Item No.		d_0 [mm]	h_2 [mm]	t_{fix} [mm]	l [mm]	$\phi \times$ length [mm]	T_{inst} [Nm]	SW [mm]	[pcs]
Item	gvz	R									
FAZ II Classic 8/10	570415	–	C1	8	68	10 / 20	77	M8 x 38	20	13	50
FAZ II Classic 8/10	–	570449	C1	8	68	10 / 20	75	M8 x 37	20	13	50
FAZ II Classic 8/30	570416	–	C1	8	88	30 / 40	97	M8 x 58	20	13	50
FAZ II Classic 8/30	–	570450	C1	8	88	30 / 40	95	M8 x 57	20	13	50
FAZ II Classic 8/50	570418	–	C1	8	108	50 / 60	117	M8 x 78	20	13	50
FAZ II Classic 8/50	–	570451	C1	8	108	50 / 60	115	M8 x 77	20	13	50
FAZ II Classic 8/100	570419	–	C1	8	158	100 / 110	167	M8 x 128	20	13	25
FAZ II Classic 8/100	–	570452	C1	8	158	100 / 110	165	M8 x 127	20	13	20
FAZ II Classic 10/10	570422	–	C1 / C2	10	85	10 / 30	97	M10 x 53	45	17	50
FAZ II Classic 10/10	–	570455	C1 / C2	10	85	10 / 30	95	M10 x 52	45	17	50
FAZ II Classic 10/20	570423	–	C1 / C2	10	95	20 / 40	107	M10 x 63	45	17	25
FAZ II Classic 10/20	–	570456	C1 / C2	10	95	20 / 40	105	M10 x 62	45	17	50
FAZ II Classic 10/30	570424	–	C1 / C2	10	105	30 / 50	117	M10 x 73	45	17	25
FAZ II Classic 10/30	–	570457	C1 / C2	10	105	30 / 50	115	M10 x 72	45	17	50
FAZ II Classic 10/50	570425	–	C1 / C2	10	125	50 / 70	137	M10 x 93	45	17	20
FAZ II Classic 10/50	–	570462	C1 / C2	10	125	50 / 70	135	M10 x 92	45	17	20
FAZ II Classic 10/70	–	570463	C1 / C2	10	145	70 / 90	155	M10 x 112	45	17	20
FAZ II Classic 10/80	570430	–	C1 / C2	10	155	80 / 100	167	M10 x 123	45	17	20
FAZ II Classic 12/10	570433	–	C1 / C2	12	99	10 / 30	112	M12 x 61	60	19	20
FAZ II Classic 12/10	–	570475	C1 / C2	12	99	10 / 30	109	M12 x 59	60	19	20
FAZ II Classic 12/20	570438	–	C1 / C2	12	109	20 / 40	122	M12 x 71	60	19	20
FAZ II Classic 12/20	–	570485	C1 / C2	12	109	20 / 40	119	M12 x 69	60	19	20
FAZ II Classic 12/30	570439	–	C1 / C2	12	119	30 / 50	132	M12 x 81	60	19	20
FAZ II Classic 12/30	–	570486	C1 / C2	12	119	30 / 50	129	M12 x 79	60	19	20
FAZ II Classic 12/50	570440	–	C1 / C2	12	139	50 / 70	152	M12 x 101	60	19	20
FAZ II Classic 12/50	–	570487	C1 / C2	12	139	50 / 70	149	M12 x 99	60	19	20
FAZ II Classic 16/5	570441	–	C1 / C2	16	113	5 / 25	130	M16 x 64	110	24	10
FAZ II Classic 16/5	–	570488	C1 / C2	16	113	5 / 25	127	M16 x 62	110	24	10
FAZ II Classic 16/25	570446	–	C1 / C2	16	133	25 / 45	150	M16 x 84	110	24	10
FAZ II Classic 16/25	–	570489	C1 / C2	16	133	25 / 45	147	M16 x 82	110	24	10

Assortment



Bolt anchor FAZ II Classic K



FAZ II Classic K

	Galvanised steel	Stainless steel	Seismic-Approval	Drill diameter	Min. drill hole depth for through fixings	Max. fixture thickness	Anchor length	Thread	Installation torque	Width across nut	Sales unit
	Item No.	Item No.		d_0 [mm]	h_2 [mm]	t_{fix} [mm]	l [mm]	$\emptyset \times \text{length}$ [mm]	T_{inst} [Nm]	SW [mm]	[pcs]
Item	gvz	R									
FAZ II Classic 8/5 K	570414	–	–	8	53	5	62	M8 x 23	20	13	50
FAZ II Classic 8/5 K	–	570447	–	8	53	5	60	M8 x 22	20	13	50
FAZ II Classic 10/10 K	570420	–	C1 / C2	10	65	10	77	M10 x 33	45	17	50
FAZ II Classic 10/10 K	–	570453	C1 / C2	10	65	10	75	M10 x 32	45	17	50
FAZ II Classic 10/20 K	570421	–	C1 / C2	10	75	20	87	M10 x 43	45	17	25
FAZ II Classic 10/20 K	–	570454	C1 / C2	10	75	20	85	M10 x 42	45	17	50
FAZ II Classic 12/10 K	570431	–	C1 / C2	12	79	10	92	M12 x 41	60	19	20
FAZ II Classic 12/10 K	–	570464	C1 / C2	12	79	10	89	M12 x 39	60	19	20
FAZ II Classic 12/20 K	570432	–	C1 / C2	12	89	20	102	M12 x 51	60	19	20
FAZ II Classic 12/20 K	–	570472	C1 / C2	12	89	20	99	M12 x 49	60	19	20

Accessories

Accessories



Filling disc FFD



FABS



FA-ST II



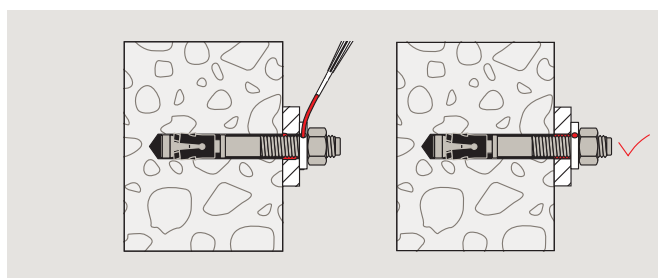
FA-ST II Set

		Internal diameter D [mm]	External-ø d [mm]	Match	Contents	Sales unit [pcs]
Item	Item No.					
FFD 26 x 12 x 6	538458	12	26	FAZ II Plus / FAZ II Classic M8/M10		4
FFD 26 x 12 x 6 R	541986	12	26	FAZ II Plus / FAZ II Classic M8/M10 R		4
FFD 30 x 14 x 6	538459	14	30	FAZ II Plus / FAZ II Classic M12		4
FFD 30 x 14 x 6 R	541987	14	30	FAZ II Plus / FAZ II Classic M12 R		4
FFD 38 x 19 x 7	538460	19	38	FAZ II Plus / FAZ II Classic M16		4
FFD 40 x 19 x 7 R	541988	19	40	FAZ II Plus / FAZ II Classic M16 R		4
FABS	077937	–	–	Bolt anchor M6 - M12		1
FA-ST II M10	558790	–	–	Bolt anchor M10	SDS Adapter; Socket SW17	1
FA-ST II M12	558791	–	–	Bolt anchor M12	SDS Adapter; Socket SW19	1
FA-ST II M16	558792	–	–	Bolt anchor M16	SDS Adapter; Socket SW24	1
FA-ST II Set	558789	–	–	Bolt anchor M10 - M16	SDS Adapter; Socket SW17, SW19, SW24	1

fischer filling disc FFD

For use under seismic loads C2 or optional to minimise the hole clearance:

The annular gap between bolt and attachment may be filled with mortar compressive strength $\geq 50 \text{ N/mm}^2$ e.g. FIS V Plus, FIS EM Plus, FIS HB and FIS SB. The backfilling disc would be used in addition to the standard washer to be used. The thickness of the backfill disc must be taken into account for t_{fix} . The counterbore in the filling disk must point in the direction of the attachment part.



Loads

Bolt anchor FAZ II Classic

Permissible loads of a single anchor¹⁾ in normal concrete of strength class C20/25.

For the design the complete current assessment ETA-23/0162 of 14.03.2023 has to be considered.

Type	Material/ surface ²⁾	Effective anchorage depth h_{ef} [mm]	Minimum member thickness h_{min} [mm]	Instal- lation torque T_{inst} [Nm]	Cracked concrete				Non-cracked concrete			
					Permissible tension (N_{perm}) and shear loads (V_{perm}); minimum spacing (s_{min}) and edge distances (c_{min}) with reduced loads				Permissible tension (N_{perm}) and shear loads (V_{perm}); minimum spacing (s_{min}) and edge distances (c_{min}) with reduced loads			
					$N_{perm}^{3)}$ [kN]	$V_{perm}^{3)}$ [kN]	$s_{min}^{3)}$ [mm]	$c_{min}^{3)}$ [mm]	$N_{perm}^{3)}$ [kN]	$V_{perm}^{3)}$ [kN]	$s_{min}^{3)}$ [mm]	$c_{min}^{3)}$ [mm]
FAZ II Classic 8	gvz	35	80	20	2.6	8.1	35	40	4.9	8.1	40	40
	gvz	90	140	20	3.8	8.1	35	40	6.7	8.1	40	40
	R	35	80	20	2.6	8.1	35	40	4.9	8.1	40	40
	R	90	140	20	3.8	8.1	35	40	6.7	8.1	40	40
FAZ II Classic 10	gvz	40	80	45	4.1	10.8	40	45	5.9	13.1	40	45
	gvz	100	150	45	6.2	13.1	40	45	9.5	13.1	40	45
	R	40	80	45	4.1	10.8	40	45	5.9	13.1	40	45
	R	100	150	45	6.2	13.1	40	45	9.5	13.1	40	45
FAZ II Classic 12	gvz	50	80	60	5.8	18.0	50	55	8.3	18.5	50	55
	gvz	125	190	60	9.5	18.5	50	55	10.5	18.5	50	55
	R	50	80	60	5.8	18.0	50	55	8.3	18.5	50	55
	R	125	190	60	9.5	18.5	50	55	10.5	18.5	50	55
FAZ II Classic 16	gvz	65	100	110	8.6	27.5	65	65	12.3	34.2	65	65
	gvz	160	240	110	12.9	34.2	65	65	18.4	34.2	65	65
	R	65	100	110	8.6	27.5	65	65	12.3	34.2	65	65
	R	160	240	110	12.9	34.2	65	65	18.4	34.2	65	65

¹⁾ Design according to EN 1992-4:2018 (for static resp. quasi-static loads). The partial safety factors for material resistance as regulated in the ETA as well as a partial safety factor for load actions of $\gamma_L = 1.4$ are considered. As a single anchor counts e.g. an anchor with a spacing $s \geq 3 \times h_{ef}$ and an edge distance $c \geq 1.5 \times h_{ef}$. For anchorage depths of less than 40 mm, the use of a single anchor is only permitted for multiple use of redundant non-structural applications. Accurate data see ETA.

²⁾ Further steel grades, versions and technical data see ETA, e.g. for dry internal conditions, galvanised steel (gvz); for damp interiors and for outdoor use, stainless steel (R).

³⁾ In the case of combinations of tension and shear loads, bending moments with reduced or minimum spacing and edge distances (anchor groups), the design must be carried out in accordance with the provisions of the complete ETA and the provisions of the EN 1992-4:2018. We recommend using our anchor design software C-FIX.